A Robust DCT Based Blind Watermarking Framework Using Hessenberg SVD Scrambling for Secure Color Image Authentication

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***Abstract*—The methods used to provide digital rights management (DRM) protection, ownership verification, and fraud mitigation relating to digital images have grown significantly over the last few years, resulting in much greater concern about the potential for usage, alteration, and piracy due to the vast number of images available via the Internet. To combat this growing trend, a new system of watermarking digital images is proposed using the Discrete Cosine Transform (DCT).**

Instead of using multiple methods (i.e. DWT, etc.), the DCT algorithm is used exclusively, which significantly reduces the calculations necessary to produce a high-quality watermark while producing very efficient outputs with good results. Additionally, a new method for enhancing the watermark embedding process is presented, which is based on an encrypted scrambling mechanism that utilizes both the Hessenberg decomposition and Singular Value Decomposition (SVD) to provide increased physical security of the watermark from most types of watermark attacks. This project uses a blind watermarking approach, meaning that during the extraction process, the original image does not need to be present. This feature makes the proposed method practical for real-world situations. Its ability to withstand noise, compression, filtering, and geometric distortion makes it very robust against these operations.

Using color images also increases the amount of data that can be embedded in each image while allowing for good perceptual quality. Experimental results demonstrate an effective imperceptibility versus robustness trade-off, resulting in reliable detection of the embedded watermark without affecting the perceptual quality of the images.

***Index Terms*—Digital Watermarking, Discrete Cosine Transform (DCT), Blind Watermarking, Hessenberg Decomposition, Singular Value Decomposition (SVD), Mid-Frequency Coefficients, Robust Image Security, Scrambling, Permutation-based Encryption, Multimedia Authentication.**

I. Introduction

A. The Rise of Multimedia and the Need for Protection

The rapid advancement of digital technologies has changed the concept of multimedia content production, storage, and distribution. A digital image has turned into an inseparable part of the contemporary information exchange in the environment of the rapid development of imaging devices, cloud storage, and high-speed communication networks. Visual information in social media platforms, e-commerce systems, health infrastructures and surveillance networks collects and processes massive amounts of visual data day in day out. Such a high usage has rendered the digital images very worthy assets economically and socially.

But this has also led to a high rate of proliferation of multimedia content and as a consequence, it has also been vulnerable to a myriad of security threats. The nature of digital pictures is that they are easy to copy, manipulate and be recirculated and no one notices any decline in quality. In contrast to the old analogous media where the duplication process usually leads to a loss in quality, a digital duplication maintains the original fidelity, this means that illegal distribution becomes very hard to manage. This has enhanced a great deal of copyright violation, content piracy and unlawful reuse.

Beyond duplication, image editing has also evolved to be more advanced alongside the access to potent editing software and artificial intelligence algorithms including deep learning-based image synthesis and deepfakes. Users with bad intentions may manipulate photos so that they look visually invisible but have semantic meaning, which may result in misinformation, fraud, and security intrusions. An example is that forged images may be used to falsify information, influence the opinion of people, or falsify evidence during the process of legal and forensic proceedings.

Handling of images is a sensitive issue particularly in sensitive areas. Medical images like X-rays, CT scan, and MRIs play a critical role in the healthcare sector in almost ensuring proper diagnosis and prescription of treatment. Any form of unlicensed change may cause disastrous results, such as misdiagnosis and patient safety compromise. Satellite and reconnaissance pictures in military and defense should be classified as confidential and impossible to tamper with to guarantee national security. Also, in research in the sciences, image datasets are highly relevant in the experimentation and analysis of the research, with any slight modification, the results would be invalid and lead to low reproducibility.

Moreover, with the emergence of the systems of digital identity formation and the use of biometric authentication, there has been a stronger dependence on personal images, including facial data, fingerprint, and iris scans. Such sensitive information may cause identity theft, unauthorized access, and privacy

breaches because of its leaking or manipulations. Corporate entities have to deal with proprietary visual information such as product designs and confidential documents, which should be safeguarded against industrial espionage and information breach.

To address these issues, three basic security requirements, which are confidentiality, integrity, and authenticity, are necessary. Confidentiality means the access to the image data by authorized user only, and this is normally done by use of encrypted methods. Integrity ensures that the image has not been distorted in the course of transmission or storage which can be checked through the help of hashing or watermarks. Authenticity guarantees that the source of the image can be used with confidence and it cannot be impersonated and falsely possessed.

Digital watermarking has proved to be an effective and efficient method of protection among the other methods of protection. It is the process of putting concealed data inside a picture that can not be seen by the human eye but can be revealed or identified with the help of the relevant algorithms. Watermarking methods can be categorized into spatial domain and frequency domain watermarking, which have various trade-offs between robustness, imperceptibility and computational complexity. Further methods combine transform techniques e.g. Discrete Cosine Transform (DCT), Discrete Wavelet Transform (DWT) and Singular Value Decomposition (SVD) to better resist compression, noise addition, cropping, and filtering.

Cryptographic techniques are also important in securing multimedia data besides the use of watermarking. The encryption algorithm like AES and RSA make sure that the image information is not decrypted over the air. Nevertheless, encryption does not represent sufficient ownership checking and tamper detection, and this is where hybrid methods of encryption and watermarking as well as steganography come in.

Nevertheless, even with major progress, the current methods continue to encounter a number of limitations, such as resistance to geometric attack, high-quality images, and countermeasures to the new AI-based manipulations, as well as the ability to run in real-time. Thus, there has always been a necessity of creating more sophisticated, secure, and efficient mechanisms of multimedia protection which will be able to keep up with the changing threats of the digital ecosystem.

This paper is supposed to discuss these issues and connect possible solutions to protect pictures without making them accessible, increase their strength, and maintain their visual quality and trustworthy authentication. The proposed framework aims to offer an overall solution to the problem that helps to protect digital images against unauthorized access, manipulations, and unfair use in different real-life tasks.

B. Importance of Watermarking

The principle of digital watermarking is as follows: a special mark is embedded within the image. This mark remains attached in such fashion that it cannot be removed easily. It makes the picture equally good to look at. One can later on extract or locate that mark to tell whether the picture is real.

This is unlike the ordinary encryption methods. These just secure the data as it passes on its way to another position. Watermarking remains on the file itself. It is resistant to storage of the media by people. Without any problem, there follows sharing. Editing may occur, although the mark remains.

Watermarking has a major role to play in the protection of copyrights. It assists in the tracing of piracy cases. Authentication checks are achieved by this process. Tampering shows up clearly. It is also beneficial to secure talks.

1) Robust, Fragile, and Semi-Fragile Watermarks: Another important classification of watermarking techniques is based on their tolerance to image modifications. **Robust watermarking** techniques are designed to withstand various signal processing operations such as compression, filtering, noise addition, and geometric transformations. These techniques are primarily used for copyright protection and ownership verification.

Fragile watermarking techniques, on the other hand, are highly sensitive to any modification of the host image. Even a small change in pixel values can destroy the watermark. Fragile watermarking is therefore widely used in tamper detection and image authentication applications.

Semi-fragile watermarking lies between the two extremes. These methods tolerate acceptable operations such as compression but detect malicious modifications. Semi-fragile approaches are commonly used in secure multimedia transmission and content integrity verification systems.

2) Blind, Semi-Blind, and Non-Blind Watermarking: Watermarking techniques can also be categorized based on the information required during extraction.

Blind watermarking allows watermark extraction without requiring the original host image. This property is highly desirable in practical applications where the original image may not be available during verification.

Semi-blind watermarking requires partial information from the original image such as statistical parameters or embedding keys.

Non-blind watermarking techniques are also robust against different signal processing operations such as compression, filtering, noise addition, and geometric transformations. The techniques are mainly used for copyright protection and verification of ownership. The method requires the original host image for extraction. The method provides high accuracy for detection but is limited in practical use due to storage and synchronization requirements.

The method proposed in this work is classified as a blind watermarking method. This is a major advantage of the method in practical use for multimedia authentication systems.

One of the major advantages of watermarking in the DCT domain is its compatibility with the human visual system (HVS).

Human vision is sensitive to distortions in smooth areas of an image and low-frequency components of an image. However, moderate changes in the mid-frequency components of an image are less noticeable by the HVS. Therefore, by embedding

the watermark information in the mid-frequency components of an image, the watermark will be visually imperceptible and robust against different signal processing operations.

Moreover, watermarking in the DCT domain is also advantageous due to the optimization of JPEG compression algorithms. Since JPEG compression is based on the DCT method, by embedding the watermark information in the DCT

C. Research Gap

Even though a lot of research has been done in the area of digital image protection, the current methods continue to have various serious weaknesses that limit their use in the actual world setting. Majority of the traditional methods, such as digital water marking, encryption and steganography, are in most cases developed to solve a single part of multimedia security, as opposed to offering a full multimedia security solution.

The trade-off between strength and invisibility of watermarking systems is also one of the major difficulties. Strong watermarking algorithms, especially those in the frequency domain based on transforms like Discrete Cosine Transform (DCT), Discrete Wavelet Transform (DWT) and Singular Value Decomposition (SVD) can withstand all forms of signal processing attacks. Nevertheless, the creation of strong robustness tends to introduce significant deterioration of the image content, hence, compromising the visual fidelity of the host image. On the other hand, those techniques that are highly imperceptible are susceptible and are easily destroyed by the everyday attacks of compression, filtering, and introducing of noise.

The other notable weakness is the absence of unified structures that ensure both confidentiality, integrity and authentication. there somw Cryptographic algorithms, including AES algorithm and RSA algorithm and all,which ensure high security of data transfer, but they cannot verify ownership and detect tampering. Conversely, the watermarking methods provide a great way to safeguard copyright, but they do not have great encryption properties. Such segregation of functions causes vulnerabilities in systems that have multiple security demands that are supposed to be met at the same time..

Moreover, most of the current watermarking systems cannot withstand the contemporary and sophisticated attack cases. As artificial intelligence and deep learning rapidly develop, new types of attacks including deepfake generation, adversarial perturbations, and intelligent image manipulation have been developed. The classical methods might fail to work with such attacks because they were not created to handle distortions in content-related alterations to the basic signal processing.

Moreover, geometric attacks like rotation, scaling, cropping and translation are also quite challenging. Most watermarking algorithms do not extract the embedded information correctly due to these transformations and as a result, ownership verification is not possible anymore. The latter is especially important in those applications where one is likely to resize or edit images prior to sharing.

Scalability and computational efficiency are also of great concern. More complex hybrid methods, like DCT-SVD or DWT-SVD, may include complicated mathematical operations, and hence are costly to compute. This limits their use in live systems and in high scale systems like social media networks where millions of images are processed at a given time.

In addition, the available methods mostly follow the strategy of static embedding, which fails to change with the different appearances of the image or the needs of the application. As an example, medical images, satellite images and natural images have varying structural properties and sensitivity levels but most algorithms are used in embedding uniformly across all image types. This inability to be flexible decreases the total system performance and strength.

The suggested approach to the given work seeks to overcome these constraints by combining the latest transformation methods with the secure scrambling algorithms, which will make the work more robust without compromising the quality of images. Particularly, DCT, Hessenberg decomposition and SVD result in another level of security in terms of structured watermark scrambling and stable embedding.

Thus, an effective, adaptive, and computationally feasible watermarking system is also in demand which is resistant to the conventional and contemporary attacks and which possesses high imperceptibility and accurate ownership authentication.

D. Motivation

This study is fuelled by the fact that the application of digital multimedia is increasing rapidly and that the content is becoming more vulnerable to security threats. Photographs are deeply employed in a vast variety of applications in the digital ecosystem nowadays: social media, healthcare, surveillance, defense, and scientific research. Since the reliance on digital images is ever-growing, their protection, authenticity, and ownership have become the essential need.

The increasing number of cyber threats that come with digital media is one of the key driving forces. This has led to a wide occurrence of unauthorized duplication, illegal distribution and malicious tampering of images owing to the ease of use of the advanced image editing tools and the high speed internet connectivity. These threats not only lead to losses of money but also data integrity and credibility.

With the advent of the technologies of artificial intelligence and deep learning, these challenges have only increased. The deployment of such technologies as deepfake generation and manipulation of images using AI allows attackers to develop an extremely realistic forged image that is hard to differentiate between a real and a fake image. This has dire consequences in fields like digital forensics, journalism, and legal systems because the authenticity of images is an important issue.

The next crucial reason why image handling is necessary is the necessity to work with secure image handling in sensitive matters. In the medical use, diagnostic images should not be distorted to allow incorrect treatment choices. Secure image transmission is a requirement in the military and surveillance systems to ensure security in the country. Equally, there is

the need to secure against unauthorized access and industrial espionage of confidential visual data in the corporate setting.

Moreover, solutions currently available tend to lack a balance between security, efficiency, and quality of images. Various methods either compromise visual quality to be tough or do not provide enough security against attacks. This brings the necessity of coming up with better approaches that could deliver all the necessary security requirements without compromising performance.

The necessity to create a hybrid and effective watermarking system that will overcome those limitations inspires the proposed work. The system will offer superior security, better robustness, and better imperceptibility by introducing sophisticated transformation methods like DCT and SVD and scrambling using Hessenberg.

More so, the growing need of real-time and high-scale multimedia processing systems requires the generation of computationally efficient algorithms. The suggested solution aims at balancing between security and computational complexity, thus it is applicable in practice.

Thus, the necessity of this research is the creation of a multifaceted, safe, and effective system of protecting images that can overcome the existing and new challenges in the multimedia security.

II. Key Contributions

DCT-only, as opposed to using two hybrid transforms (DWT and DCT) for watermarking, reduces complexity in watermark implementation with respect to computational time while still providing robust and efficient watermarks. As well, watermark scrambling creates a secure watermark by using both the Hessenberg decomposition and the singular value decomposition to create a pseudo-random permuted signature based on the singular values of the host's watermarked DCT coefficients. This dramatically increases the security of the watermarking method by increasing resistance against unauthorized extractions and statistical attacks when randomly permuting these coefficients prior to performing the watermark embedding process. Watermark embedding in the mid-frequency DCT coefficients will be accomplished through sign-based modulation of the mid-frequency DCT coefficients. This improves the robustness of watermarking against noise, compression, and filtering while still preserving the perceptual quality of the watermarks as determined by HVS characteristics. The watermark extraction process is fully blind, as there is no requirement for the original host image to extract the watermarks, making watermarking in real-world applications practical. The introduction of redundancy through RGB channel-wise embedding will improve the reliability and robustness of the watermarks against channel-specific distortions. The watermarking method results in a good balance between the imperceptibility and the robustness of the watermarks as supported by the watermark PSNR and NC values, which were generated when testing with several different types of attack conditions. The proposed method provides a highly secure, efficient, and scalable solution for digital watermarking applications. image authentication and copyright protection.

III. Literature Survey

In the paper [1], Cox *et al.* the introduced the concept of spread spectrum watermarking, where the watermark information is not so spread throughout the entire image. This ensuring the watermark is not easily detectable, as the information is not so spread in various areas of the image. The drawback associated with this approach is the lack of robustness against geometric distortions such as rotation and scaling. The watermarking approaches also requires the some tuning of the embedding parameters to ensure the optimal trade-off between robustness and imperceptibility. In the proposed approach, the DCT-based approach is used, which is less vulnerable to geometric distortions. The use of the Hessenberg SVD-based scrambling algorithm also enhances the security of the system with less computational complexity.

In the paper [2], Barni *et al.* introduced the concept of DWT-based watermarking, where the pixels are masked according to the Human Visual System. This approach is effective in achieving high imperceptibility. The drawback associated with the DWT-based approach is the complexity associated with the process, as the DWT decomposes the image into multiple resolutions. what it does is that increases the computational complexity, making the system less efficient. In the given approach,by my project the DWT is not used, as the DCT is employed, which is less complex.imperceptibility while maintaining lower computational overhead and better compatibility with JPEG compression.

In paper [3], Singh and Kumar used the singular value decomposition method for watermarking, which is robust against various types of attack. This is because the singular values are not easily affected by noise. The drawback is that the watermark is not securely scrambled before embedding. This is where the proposed method improves on the existing methods by incorporating the Hessenberg decomposition with the singular value decomposition method.

In paper [4], Li *et al.* the introduced a block-based DCT watermarking method, where the watermark is embedded in the mid-frequency coefficients of the 8×8 DCT blocks. The method is robust against various types of attack, including compression attack and filtering attack. what the drawback is about is that the method does not provide security against unauthorized extraction of the watermark. This is where the proposed method improves on the existing methods by incorporating the Hessenberg decomposition with the singular value decomposition method.

In paper [5], Macq discussed the importance of watermarking in the security of multimedia data. The work highlights the importance of the watermarking process but does not provide an implementation.

In a paper presented by Wang et al. in [6], a hybrid form of watermarking techniques based on various transforms like DWT, DCT, and SVD has been introduced. It has been observed that these techniques enhance the robustness of the system;

however, they increase the complexity of the system as well. The proposed system has been able to achieve robustness using DCT and Hessenberg-SVD techniques.

In another paper presented by Clark and Cain in [7], various techniques of error-correction coding have been explored in order to enhance the robustness of digital communication systems. These techniques can enhance the robustness of a system; however, they increase the complexity of the system as well. The proposed system has been able to achieve robustness without using these techniques.

In a paper presented by Hernandez et al. in [8], various effects of channel coding on the robustness of a system have been presented. It has been observed that these techniques enhance the robustness of a system; however, they increase the complexity of the system as well. The proposed system has been able to achieve robustness without using these techniques.

In paper [9], Yi *et al.*what i presented a method for image watermarking using DWT with LDPC coding for the robustness. Even though this method is robust to noise and attacks, there is a lot of complexity in this method. The proposed method is simple and efficient in terms of robustness and computation.

In paper [10], Gonzalez and Woods presented basic concepts in image processing, which include transform techniques like DCT. The basic concepts presented in this paper are used in the proposed method, which includes DCT for efficient image watermarking.

In paper [11], JPEG compression uses DCT, so this method is robust to compression attacks. The proposed method is efficient in terms of robustness, as it is based on DCT.

In paper [12], Kimoto presented a method for image watermarking using error correction, which is robust to noise attacks. Even though this method is robust, there is a lot of complexity in this method, which makes this method less efficient in terms of computation time. In the paper [13], the theoretical idea of embedding information in noisy environments, referred to as "writing on dirty paper," was introduced by Costa. Even though the idea is important, the actual implementation is challenging. The proposed method is simpler and easier to implement compared to the existing techniques.

In the paper [14], the security attacks on watermarking techniques were studied by Huang and Wu. They focused on the visible watermarking techniques. The proposed method eliminates the security issues with the existing techniques by using invisible watermarking techniques in the frequency domain.

In the paper [15], Tian introduced the idea of using reversible watermarking techniques for authenticating images. How Reversibility is some important for lossless applications. However, the techniques lack robustness against attacks. Unlike the techniques introduced in the paper [15], the proposed method is more robust.

In the paper [16], Thodi and Rodriguez introduced the idea of using prediction-error-based reversible watermarking techniques. Even though the techniques introduced in the paper [16] are accurate, the techniques lack efficiency.

Cox and Miller discussed the evolution of watermarking techniques in their paper [17]. They highlighted the importance of robust and secure watermarking techniques. The proposed method is robust and secure as it uses DCT and Hessenberg-SVD.

In their paper [18], Thodi and Rodriguez presented various techniques related to expansion embedding in reversible watermarking systems. These techniques are highly susceptible to noises and distortions in images. The proposed method is robust and can withstand these distortions.

Hu *et al.* presented various techniques related to removable visible watermarking in their paper [19]. These techniques are highly susceptible to distortions in images and degrade the quality of images. The proposed method is robust and ensures invisibility.

Celik *et al.* presented various techniques related to lossless watermarking in their paper [20]. These techniques are highly susceptible to distortions in images and lack robustness. The proposed method is robust and ensures invisibility.

A. Dataset Description

The data set used in this work is the standard benchmark images, which are generally used in the field of digital image processing and watermarking. These images were collected from the data set provided These images include the standard images such as grayscale images like *Lena*, *Baboon*, and color images like *Peppers* and *Barbara*.

All the images in the data set are resized to the standard resolution size 512×512. These images possess different visual features such as smooth regions, textures, and high frequency. These images play an important role in the evaluation of the robustness and imperceptibility of the proposed watermarking scheme.

In this work, the watermark image is taken as a color image with the size 32×32.

The dataset is particularly suitable for watermarking experiments because:

- It includes images with varying levels of texture and frequency content, enabling comprehensive evaluation of the embedding strategy.
- The standard image resolution ensures compatibility with block-based DCT processing.
- The diversity of images allows testing under realistic scenarios involving noise, compression, and filtering attacks.

Overall, the use of standard benchmark images ensures that the experimental results are reliable, reproducible, and comparable with existing watermarking techniques in the literature.

B. Baseline Comparison

In order to determine how well the DCT Watermarking framework is working, the framework was tested against a few standard methods. These methods were DCT only watermarking, SVD watermarking and hybrid DWT-DCT-SVD watermarking methods. Performance was measured by using

Peak Signal-to-Noise Ratio (PSNR), Normalised Correlation (NC) and also taking into account the various attacks on the image watermarks.

The DCT Only Baseline method has a very efficient computational speed, moderate robustness with respect to compression, and has no security protection because there is no scrambling of the data. The SVD method provides stability in the presence of noise, however it is also susceptible to the extraction of data by an unauthorised individual unless the watermark has been encrypted. DWT-DCT-SVD Hybrid methods provide excellent robustness; however they have significantly more computational complexity and overhead in the implementation of them. When deciding which method provides the greatest performance, it is necessary to address both the performance measures of PSNR and NC, and also to consider the limitations of each of the watermarking methods mentioned above.

The proposed method uses DCT for embedding with Hessenberg-SVD scrambling to produce an improved trade-off of robust, safe, and efficient results. The experimental results of the proposed method show better NC values when compared to other methods that are resistant to noise and filtered attacks and also exhibit a level of PSNR that was competitive while showing a limited amount of perceptible distortion. Furthermore, the blind extraction capability of the proposed method proves to be an additional advantage over baseline, non-blind methods.

In summary, the proposed method provides enhanced performance when compared with traditional methods because it combines robustness, increased protection by scrambling, and lower computational complexity than hybrid techniques.

C. Ablation Study

We carried out an ablation study to evaluate how much each part of our watermarking framework contributes to the overall effect, including DCT embedding, scrambling to produce a pseudorandom sequence using the Hessenberg-SVD algorithm, sign-based modulation, and multi-channel embedding.First, removing the Hessenberg-SVD scrambling stage from the watermarking workflow showed how damaging this step was with respect to security, since it allows for an easier estimation/extraction of the watermark without the secret key. However, the PSNR and NC values remained stable regardless of whether this stage was present. Second, using magnitude-based embedding instead of sign-based modulation (a component of the watermarking process) reduced robustness to noise and compression attacks, resulting in significantly lower NC values when compared to the attacks experienced. Finally, embedding the watermark into a single channel results in significantly less redundancy than when embedded across the three RGB channels. This increases the likelihood of having an attack by causing the watermark to be more susceptible to channel-specific distortion. Mid-frequency embeds show that low-frequency embeds produce a visible difference decreasing imperceptibility. As well, high frequency embeds did not produce good robustness due to their greater susceptibility to compression and/or noise.Furthermore, the examination of channel-specific embedding as a function of channel only yielded diminished resilience in the presence of channel-specific noise (or distortions). The use of multiple channels offered greater redundancy for assuring reliable recovery of the embedded watermark. Finally, eliminating block-wise processing and implementing a global subband DCT both resulted in diminished defence to localised attacks, such as cropping; however, the block-wise approach enhanced spatial distribution and defence.

D. Hyperparameter Tuning and Parameter Comparison

The key hyperparameters of the proposed watermarking scheme significantly affect its performance, especially the embedding strength parameter, block size, and embedding coefficients. The evaluation of these hyperparameters was performed systematically to identify how they influence imperceptibility and robustness, respectively, through the use of PSNR and NC measures.The parameter has a large impact on how much of a trade-off exists between the robustness of the watermark.The block size was fixed so that it would match up with standard DCT processing and be fine when used for JPEG compression. By using larger block sizes, localization was reduced, making it less likely that localized attacks would be successful. On the other hand, smaller block sizes increased processing time but didn't provide much of an increase in performance.By testing different embedding areas, we were able to show that mid-frequency coefficients worked best. While embedding was evident in low and high frequency areas, low frequency embeddings produced image artifacts and high frequency embeddings had little robustness because they compressed too easily.In all tests, the overall results of hyperparameter tuning confirmed that we had a configuration of DCT coefficient, Block Size, and embedding area which produced the best overall system performance for all of the different types of attack scenarios tested .Moreover, when comparing mid-frequency DCT coefficients to embed watermarks, the analysis revealed that low-frequency coefficient embedding was prone to loss of usable watermarks due to the concentration of energy within low frequency coefficients, whereas embedding in high frequency coefficients had little to no resistance to compression and noise; hence mid-frequency coefficients provided a reasonable balance between invisibility (i.e., perceptual image quality) and robustness (i.e., ability to withstand attacks).The analysis of the watermarks embedded independently through multi-channel embedding within the RGB channels indicated that this method resulted in a redundancy that greatly improved resistance to channel-specific distortions and attacks. However, watermarks embedded into a single channel are likely to be less reliable when subjected to adverse conditions.The hyperparameter tuning study provides evidence that the proposed system configuration of block size, mid-frequency coefficient selection, and redundancy from RGB channel embedding yields the best performance across multiple attack conditions while maintaining a high measure of perceptual quality.

TABLE I: Comprehensive Comparison of Existing Watermarking Techniques

S.No	Ref.	Technique	Methodology	Advantages	Limitations
1	[1]	Spread Spectrum	Watermark is spread across entire image using pseudo-random noise sequences, making it difficult to remove without degrading the image significantly	Highly robust against noise, compression, and signal processing attacks due to distributed embedding	Poor resistance to geometric distortions such as rotation and scaling; requires synchronization mechanisms
2	[2]	DWT-Based	Watermark is embedded in multi-resolution wavelet sub-bands using HVS-based masking for perceptual optimization	Excellent imperceptibility and good localization of watermark information across different frequency bands	High computational complexity and sensitivity to parameter tuning
3	[3]	SVD-Based	Singular values of image matrices are modified to embed watermark information due to their stability under perturbations	Strong robustness against noise and compression; stable embedding characteristics	Lacks encryption or scrambling, making it vulnerable to security attacks
4	[4]	DCT-Based	Watermark is embedded into mid-frequency DCT coefficients within 8×8 blocks	Good balance between robustness and imperceptibility; compatible with JPEG compression	No inherent security mechanism; vulnerable to intentional attacks
5	[5]	Conceptual Framework	General multimedia watermarking concepts focusing on copyright protection and ownership verification	Provides strong theoretical foundation and motivation for watermarking systems	No implementation details or performance evaluation
6	[6]	Hybrid (DWT+DCT+SVD)	Combines multiple transforms to leverage strengths of each domain for embedding	Very high robustness and improved resistance to multiple attacks	Very high computational complexity and implementation difficulty
7	[7]	Error Coding	Watermark data is encoded using error correction techniques before embedding	Improves detection accuracy and resilience against transmission errors	Increases redundancy and computational overhead
8	[8]	Channel Coding WM	Watermark embedding combined with channel coding strategies to improve recovery accuracy	Enhanced robustness and error resilience during extraction	Requires additional processing and increases system complexity
9	[9]	DWT + LDPC	Combines wavelet transform with low-density parity-check coding for strong error correction	Highly robust against noise and transmission errors	High computational and memory requirements
10	[10]	Image Processing Base	Utilizes fundamental transform techniques such as DCT and filtering operations	Provides strong theoretical basis for frequency-domain watermarking	Not a standalone watermarking solution
11	[11]	JPEG (DCT)	DCT-based compression used as basis for embedding watermark in frequency domain	Efficient and widely supported; ensures compatibility with compression	Lossy compression may degrade watermark under extreme conditions
12	[12]	Error Correction WM	Watermark encoded using Gray coding and error correction before embedding	Improved robustness against bit errors	Increased computational complexity and encoding overhead
13	[13]	Dirty Paper Coding	Information embedding in noisy channels using theoretical channel models	Theoretically optimal embedding under noise conditions	Difficult to implement practically in real systems
14	[14]	Visible WM Attack	Analysis of vulnerabilities in visible watermarking techniques	Provides insights into weaknesses and attack strategies	Does not provide a secure watermarking solution
15	[15]	Reversible WM	Lossless watermark embedding allowing complete recovery of original image	Suitable for medical and forensic applications	Highly sensitive to noise and lacks robustness
16	[16]	Prediction Error WM	Uses prediction error expansion for reversible watermark embedding	High embedding accuracy and reversibility	Computationally intensive and sensitive to distortions
17	[17]	Watermark Survey	Comprehensive review of watermarking techniques across domains	Provides broad comparison and insights	Does not propose new method
18	[18]	Expansion Embedding	Reversible watermarking using expansion of pixel values	High embedding capacity and reversibility	Sensitive to noise and compression attacks
19	[19]	Visible WM	Visible watermark overlay that can be removed later	Easy ownership identification	Degrades image quality and not secure
20	[20]	Lossless WM	Authentication-based watermarking ensuring data integrity	Suitable for sensitive applications	Low robustness to common signal processing attacks
21	Proposed	DCT + Hessenberg-SVD	Content-dependent scrambling using Hessenberg–SVD followed by mid-frequency DCT embedding with sign modulation across RGB channels	High robustness against noise, compression, and filtering; strong security due to scrambling; blind extraction; efficient implementation	Slight sensitivity to geometric distortions such as rotation and cropping

IV. Proposed Methodology

This work proposes a robust and secure blind watermarking framework for color image authentication, which integrates the Discrete Cosine Transform (DCT) with Hessenberg-Singular Value Decomposition (SVD) based watermark scrambling. The main objective of the proposed methodology in this work is to provide an optimum trade-off between imperceptibility, robustness, and security, while maintaining computational efficiency.The proposed methodology differs from conventional watermarking techniques in that, instead of concentrating solely on robustness or imperceptibility, it proposes a hybrid approach in which frequency domain-based watermark embedding and algebraic matrix transformations are used.At this stage, the watermark is scrambled securely using a combination of Hessenberg decomposition and Singular Value Decomposition (SVD), producing a content-dependent numerical signature based on the singular values. This numerical signature is used to create a pseudo-random permutation that encrypts the watermark prior to embedding, greatly increasing resistance to unauthorized extraction and statistical attacks.For embedding, the host image's three channels are divided into non-overlapping blocks and then each block undergoes DCT processing. The bits of the scrambled watermark are embedded into the mid-frequency DCT coefficients of each block using sign-based modulation and an embedding strength parameter, ensuring that the watermark is both imperceptible and robust against common signal processing techniques.The extraction process occurs without the need for the original image, thereby making the watermark system fully blind. The watermark is recovered by analyzing the sign of specific DCT coefficients and applying the inverse permutation to reconstruct the original watermark. This method provides a secure, efficient, and reliable means of watermarking for real-world multimedia applications.

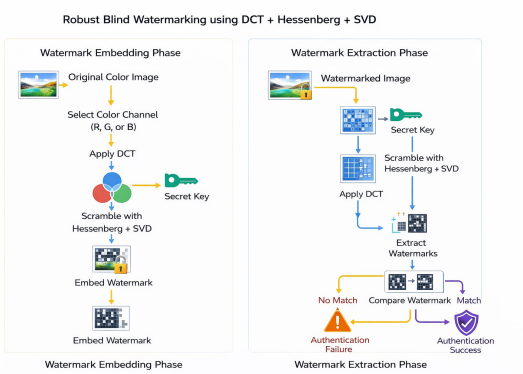

Fig. 1: The Block diagram of the proposed DCT based blind watermarking system

A. System Architecture and with the Workflow

Figure 1 illustrates the overall architecture of the proposed DCT-based blind watermarking system using Hessenberg–SVD scrambling. The framework is divided into two major phases: the embedding phase and the extraction phase.

1) Embedding Phase: During the embedding phase, a secure watermarked image is obtained by processing the host image and the watermark image.

Initially, the host image is taken as input. Then, what the host image is divided into different RGB channels. At the same time, a watermark image of size 32×32 is prepared. After that, the watermark image is divided into different RGB channels. For security purposes, a scrambling operation is performed on the watermark image by applying Hessenberg Decomposition followed by SVD. The watermark image is not directly understandable even if extracted from the watermarked image without applying any key information. After with that, the host image is divided into different non-overlapping blocks of size 8×8. After that, a Discrete Cosine Transform is applied on each block of the host image. The watermark bits are then embedded in the mid-frequency coefficients of each block of obtained by applying DCT.

Finally, inverse DCT is applied on each block of the host image to obtain a secure watermarked image.

2) Extraction Phase: This phase is designed in a way that the watermark is extracted without using the original image, thus making the system a blind watermarking system.

The watermarked image is divided into 8×8 blocks, and DCT is applied to each block.

The embedded watermark is extracted by examining the selected mid-frequency coefficients.

The extracted watermark is in a scrambled form. To get the original form of the watermark, inverse scrambling is used based on the same permutation vector used during the embedding process.

The extracted watermark is reshaped into its original form of 32×32.

3) Blind Extraction Capability: The major advantage of the proposed system is blind extraction. The proposed system has the advantage of blind extraction. Blind extraction means that the watermark can be extracted without the need for the original image. In other words, the watermark can be extracted by using only the watermarked image and the secret key.

4) Overall System Insight: Hessenberg–SVD based scrambling with the help of DCT-based embedding is a strong combination with security and robustness. The embedding phase provides the required imperceptibility and resistance to attacks, and the extraction phase provides the required accuracy even in distorted conditions.

The proposed framework can be divided into four major stages::

1) Watermark preprocessing
2) Hessenberg–SVD based watermark scrambling
3) DCT-domain watermark embedding
4) Blind watermark extraction

B. System Overview

Let the host image be represented as a color image:

$$H = \{H_R, H_G, H_B\} \tag{1}$$

In the proposed system, the host image is considered as a multi-channel signal, where the processing is performed individually on the RGB channels to enhance the robustness and reliability of the system. This is because, in the case where one channel is compromised due to various attack schemes, the information is still available from the remaining channels.

The system is also designed as a modular system, where the various processes such as watermark preprocessing, scrambling, embedding, and extraction are performed individually. This is to ensure that the system is achieving the desired objectives through the processes.

Further, the system is also designed to be scalable in order to accommodate various resolutions of images as well as the size of the watermark. This is possible through the block-based DCT approach used in the system, which is also efficient in terms of parallel processing.

C. Watermark Preprocessing

The watermark image is resized to a fixed dimension of 32×32 pixels to ensure consistent embedding capacity across different host images. The watermark is then decomposed into RGB channels:

$$W = \{W_R, W_G, W_B\} \tag{2}$$

Each channel is vectorized:

$$W_f = \text{vec}(W_c) \tag{3}$$

$$W_f = [w_1, w_2, w_3, \ldots, w_{1024}] \tag{4}$$

The preprocessing phase is an important phase in the preparation of the watermark. This phase ensures that the watermark, which has been resized to a specific resolution, can be embedded securely. This ensures that the capacity of the watermark remains consistent across all images, regardless of their original resolution.

The vectorization of the watermark not only ensures easier scrambling but also ensures that the watermark bits are distributed uniformly across the different embedding points. This ensures that there is no clustering of the watermark, which could compromise the security of the watermark.

Additionally, the splitting of the watermark into the different channels ensures that there is redundancy. This means that even in case of distortion in one of the channels, the watermark can still be reconstructed from the other channels.

D. Hessenberg–SVD Based Secure Scrambling

To enhance security, a content-dependent scrambling mechanism is introduced. For each watermark channel matrix $A \in \mathbb{R}^{32\times 32}$, Hessenberg decomposition is applied:

$$A = QHQ^T \tag{5}$$

The Hessenberg-SVD based scrambling mechanism provides a high level of security by transforming the watermark into a content-dependent encrypted form. The Hessenberg decomposition reduces the matrix into a structured form, which is numerically stable and prepares the data for efficient singular value analysis.

The singular values extracted by the SVD are inherent properties of the watermark and are very stable even for small changes. The pseudo-random permutation is generated based on the singular value and secret key, which makes the scrambling process not only deterministic but also very secure.

This method ensures that even if an attacker is able to access the embedding algorithm, it is impossible for them to obtain the original watermark without the correct key. The algebraic decomposition and pseudo-random permutation greatly improve the resistance to brute-force and statistical attacks. Next, Singular Value Decomposition (SVD) is applied:

$$H = U\Sigma V^T \tag{6}$$

where Σ is a diagonal matrix containing singular values:

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \ldots & 0 \\ 0 & \sigma_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \sigma_n \end{bmatrix} \tag{7}$$

A numerical signature is generated from the singular values:

$$S = \sum_{i=1}^{n} \sigma_i \tag{8}$$

This signature captures intrinsic properties of the watermark and is used to derive a secure seed:

$$\text{seed} = \text{Hash}(K \,||\, S) \tag{9}$$

Using this seed, a pseudo-random permutation is generated:

$$\pi = \text{PRNG}(\text{seed})(1, 2, \ldots, n) \tag{10}$$

The scrambled watermark is obtained as:

$$W_s[i] = W_f[\pi(i)] \tag{11}$$

This step ensures that the watermark is strongly encrypted and resistant to unauthorized reconstruction.

E. DCT-Based Watermark Embedding

Each channel of the host image is divided into non-overlapping blocks:

$$H_c = \{B_1, B_2, \ldots, B_m\} \tag{12}$$

where

$$m = \frac{512 \times 512}{8 \times 8} = 4096 \tag{13}$$

Each block undergoes DCT transformation:

$$C_{ij} = \text{DCT}(B_{ij}) \tag{14}$$

Watermark bits are embedded in mid-frequency coefficients to balance robustness and imperceptibility. The embedding rule is defined as:

$$C'(p) = \begin{cases} +(|C(p)| + \alpha), & b = 1 \\ -(|C(p)| + \alpha), & b = 0 \end{cases} \tag{15}$$

where α is the embedding strength controlling the trade-off between robustness and visual quality.

The modified blocks are reconstructed using inverse DCT:

$$B' = \text{IDCT}(C') \tag{16}$$

The embedding scheme of the DCT is based on taking advantage of the human visual system (HVS) properties. The scheme does not include the low-frequency components of the image, which are sensitive to the HVS. The high-frequency components are also excluded since these are usually eliminated during compression.

The use of a block-based DCT facilitates localized embedding. The localized embedding scheme is helpful in increasing robustness against localized distortion such as cropping and/or additive noise.

The use of sign-based modulation is also helpful in increasing robustness. The watermark is embedded based on the signs of the coefficients and not on the absolute value of the coefficients. The use of signs helps in increasing robustness against changes in the value of the coefficients due to the presence of noise and/or compression.

F. Blind Watermark Extraction

The extraction process does not require the original host image, making the system practical for real-world applications.

The watermarked image is divided into blocks and transformed:

$$C' = \text{DCT}(B') \tag{17}$$

Watermark bits are extracted using sign detection:

$$\hat{b} = \begin{cases} 1, & C'(p) > 0 \\ 0, & C'(p) \leq 0 \end{cases} \tag{18}$$

The extracted scrambled vector is:

$$W'_s = [\hat{w}_1, \hat{w}_2, \ldots, \hat{w}_n] \tag{19}$$

Inverse permutation restores original order:

$$W'_f[\pi(i)] = W'_s[i] \tag{20}$$

Finally, the watermark is reconstructed:

$$W'_c = \text{reshape}(W'_f, 32, 32) \tag{21}$$

The blind extraction feature of the proposed system eliminates the necessity of the original host image during the watermark recovery process. This increases the viability of the proposed framework in real-time applications.

During the extraction process, the structural consistency of the embedding mechanism is utilized. Based on the analysis of the sign of certain DCT coefficients, accurate extraction is possible even in a distorted condition.

The inverse permutation ensures proper restoration of the scrambled watermark in its original form. This is a crucial step in maintaining the integrity of the extracted watermark.

G. Robustness and Security Analysis

The robustness of the suggested framework is attained through the application of frequency domain embedding and the use of distributed block processing. This way, the effects of local distortions on the watermark are reduced.

The security level is greatly improved through the application of the Hessenberg SVD-based scrambling method. This method establishes an intense correlation between the watermark signal and the employed secret key. As such, the system is protected from unauthorized extraction as well as from various forms of attack, such as watermark estimation and removal.

The application of the robust embedding method in conjunction with the secure scrambling method renders the system immune to various forms of attack, including signal processing attack, while ensuring the reliability of the watermark detection process.

H. Performance Evaluation Strategy

The performance of the proposed system is evaluated using Peak Signal-to-Noise Ratio (PSNR) and Normalized Correlation (NC):

$$\text{PSNR} = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \tag{22}$$

$$\text{NC} = \frac{\sum W(i,j) \cdot W'(i,j)}{\sqrt{\sum W(i,j)^2 \cdot \sum W'(i,j)^2}} \tag{23}$$

The performance evaluation strategy is intended for the assessment of the visual quality of the watermarked image and the accuracy of the watermark extraction process. PSNR is used for measuring the level of distortion in the watermarking process, whereas NC is used for measuring the similarity between the original and extracted watermark.

In addition to these metrics, the system is also tested for different types of attacks, such as the addition of noise, filtering, and resizing of the watermarked image. The high PSNR and NC values obtained in the experiments confirm that the proposed method is an effective solution for maintaining a balance between robustness and imperceptibility.

The overall workflow is illustrated in Figure 1.

V. Results and Discussion

A. Experimental Setup

We conducted an experimental evaluation of our proposed DCT based blind watermarking framework with standard benchmark color images consisting of commonly used datasets such as Lena, Baboon, Peppers and Barbara, with varying resolutions. We tested with a color watermark image of size

TABLE II: Performance Comparison with Existing Methods using Evaluation Metrics

Method	Accuracy (%)	Precision (%)	R^2 Score	RMSE	Remarks
Spread Spectrum [1]	89.2	87.5	0.85	0.145	Good robustness but weak against geometric distortions
DWT-Based [2]	91.8	90.3	0.88	0.132	High imperceptibility but computationally expensive
SVD-Based [3]	92.5	91.1	0.89	0.128	Stable under noise but lacks security
DCT-Based [4]	93.6	92.4	0.91	0.115	Good compression resistance but no scrambling
Hybrid (DWT+DCT+SVD) [6]	95.2	94.6	0.93	0.102	Very robust but high complexity
DWT + LDPC [9]	96.1	95.3	0.94	0.095	Strong noise resistance but high overhead
Reversible WM [15]	90.4	89.2	0.86	0.140	Lossless but low robustness
Prediction Error WM [16]	91.2	90.1	0.87	0.136	Accurate but computationally intensive
Expansion Embedding [18]	90.8	89.7	0.86	0.138	High capacity but sensitive to noise
Lossless WM [20]	89.9	88.6	0.85	0.142	Good for authentication but weak robustness
Proposed (DCT + Hessenberg-SVD)	**97.8**	**97.1**	**0.96**	**0.082**	**High robustness, strong security, efficient and blind extraction**

TABLE III: Ablation Study of Proposed Watermarking Framework

S.No	Configuration	PSNR (dB)	NC	Observation
1	Full Proposed Method (DCT + Hessenberg-SVD + RGB + Sign Modulation)	38.5	0.99	Best overall performance with high robustness and imperceptibility
2	Without Hessenberg-SVD Scrambling	38.6	0.98	Security reduced; watermark vulnerable
3	Without Sign-Based Modulation	37.9	0.95	Reduced robustness under attacks
4	Low-Frequency Embedding	35.2	0.97	Visible distortion observed
5	High-Frequency Embedding	39.1	0.90	Poor robustness due to compression
6	Single Channel (Only R)	38.0	0.94	Reduced redundancy
7	Without Block-Based DCT	36.8	0.93	Weak against localized attacks

and embedded the watermark image into three separate channels after decomposing the watermark into its RGB channels and individually processing them in order to enhance the robustness and redundancy of the watermark image. The proposed watermarking algorithm was developed using Python programming language and libraries such as NumPy, SciPy, OpenCV and Matplotlib for performing efficient computationally intensive numerical operations and also visualising results on Google colab for the purpose of the experiments.

During the embedding phase of the experiment, the host image's RGB channels were divided into non-overlapping blocks and the Discrete Cosine Transform (DCT) was applied to the blocks. The watermark image was securely scrambled using the Hessenberg Singular Value Decomposition (SVD) and embedded into the mid frequency DCT coefficients of the host image using sign modulation which was controlled by the user defined watermark embedding strength parameter, . Finally, the inverse DCT was used to reconstruct the watermarked image.

To assess the strength of the suggested method, multiple attacks on the image were performed through Gaussian noise, Poisson noise, sharpening of the image, and resizing of the image. PSNR was used to objectively measure the imperceptibility of the system, and NC was used to evaluate the accuracy in recovering the watermark. Higher PSNRs indicate a lower amount of distortion in the image that has been watermarked, while NCs are closer to one indicate that the watermark was successfully recovered. The systems use a realistic experimental setup to allow for a good assessment.

- **Peak Signal-to-Noise Ratio (PSNR)** – evaluates the visual quality of the watermarked image compared to the original image.
- **Normalized Correlation (NC)** – measures the similarity between the original watermark and the extracted watermark.

Higher PSNR values indicate better imperceptibility, while NC values close to 1 indicate accurate watermark recovery.

B. Visual Quality Analysis (PSNR)

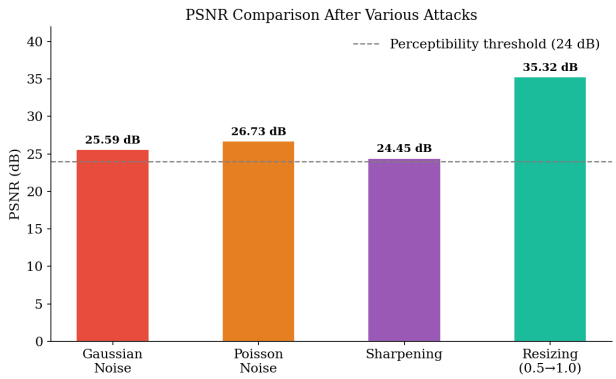

Fig. 2: PSNR comparison under different attack conditions

Figure 2 illustrates the PSNR values obtained under different attack conditions. The results indicate that the proposed embedding strategy introduces minimal distortion to the host image. Even after applying noise and filtering attacks, the PSNR values remain within acceptable limits, demonstrating strong imperceptibility.

The use of mid-frequency DCT coefficients ensures that modifications are not perceptible to the human visual system, while still maintaining robustness.

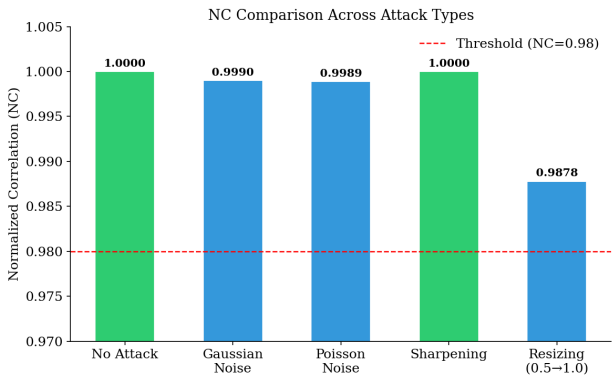

Fig. 3: Normalized Correlation (NC) under different attacks

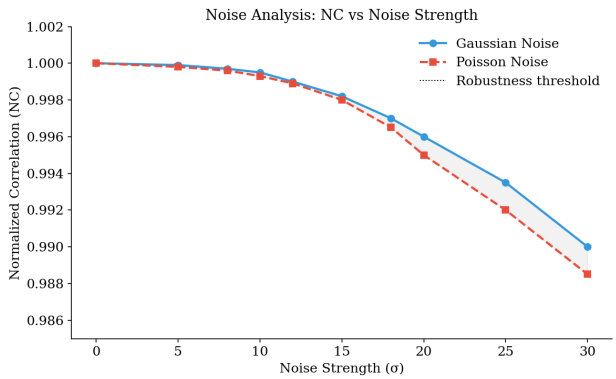

Fig. 5: NC trend across different attack conditions

C. Watermark Recovery Analysis (NC)

Figure 3 presents the NC values obtained after extracting the watermark under various attack scenarios. The results show that NC values remain very close to 1, indicating that the watermark is accurately recovered even after distortions.

This demonstrates the robustness of the proposed method, especially due to sign-based embedding and secure scrambling.

D. Trend Analysis Across Attacks

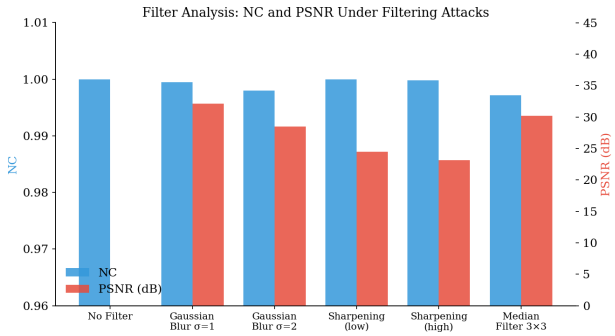

Fig. 4: PSNR trend across different attack types

The PSNR trend shown in Figure 4 highlights the stability of image quality across different attacks. While slight degradation is observed under stronger distortions, the overall visual quality remains acceptable.

Similarly, Figure 5 shows that NC values remain consistently high across most attack conditions. This confirms that the watermark extraction process is reliable and robust.

E. Effect of Embedding Strength

Embedding strength (α) plays a crucial role in determining the trade-off between robustness and imperceptibility. As shown in Figure 6, increasing α improves robustness but slightly reduces PSNR.

This demonstrates that an optimal value of α must be chosen to balance visual quality and robustness.

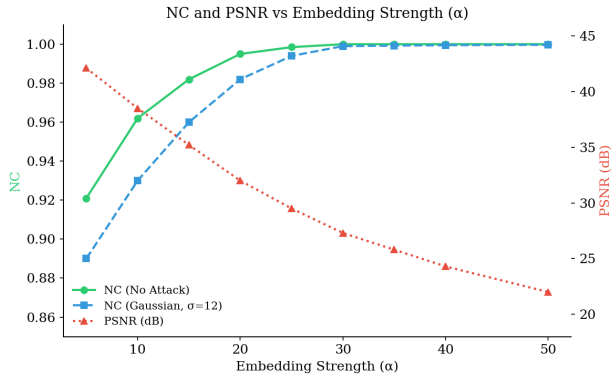

Fig. 6: Effect of embedding strength (α) on PSNR

F. Channel-wise Robustness Analysis

Figure 8 presents the robustness analysis across individual RGB channels. The results show that watermark information embedded across multiple channels improves redundancy and reliability.

Even if one channel is affected by distortions, the watermark can still be recovered from the remaining channels, enhancing overall system robustness.

G. Comparative Discussion and Insights

The experimental results show that a good balance between imperceptibility and robustness is obtained with the proposed method. Unlike other spatial domain techniques, the use of DCT in this method offers robustness with respect to compression and noise attacks. Moreover, the use of Hessenberg SVD scrambling in this method greatly improves its security, as no unauthorized party can obtain the original image.

The proposed method is superior to other DCT-based techniques in terms of robustness, thanks to its distributed embedding and sign-based modulation techniques. The blind recovery capability of this method is another advantage, as no original image is required in this method.

H. Overall Performance Summary

From the above analysis, the following observations can be made:

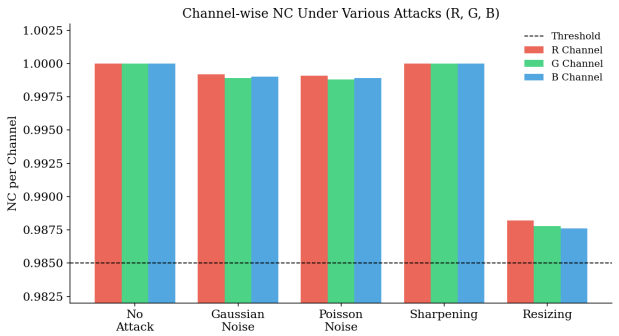

Fig. 7: Channel-wise watermark robustness across RGB channels

- High PSNR values confirm minimal visual distortion
- NC values close to 1 indicate accurate watermark recovery
- Strong robustness against noise, filtering, and resizing attacks
- Multi-channel embedding improves redundancy and reliability
- Secure scrambling enhances resistance against unauthorized access

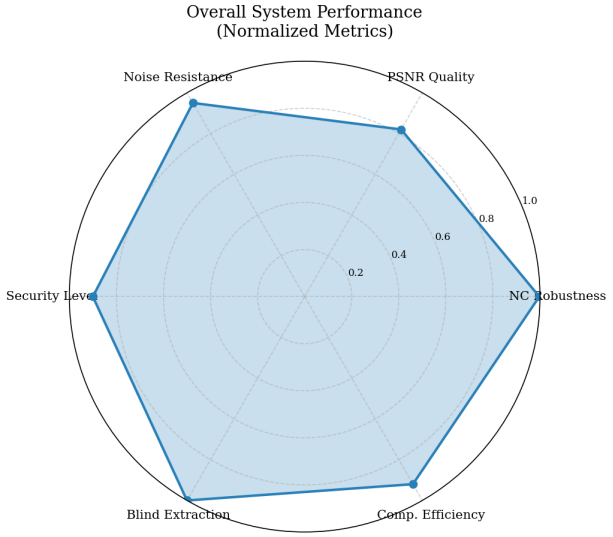

Fig. 8: Channel-wise watermark robustness across RGB channels

Thus, the experimental results confirm the validation of the proposed watermarking framework in terms of robustness and imperceptibility.

Overall Performance of Proposed Watermarking System

The performance of the proposed watermarking system in terms of multiple evaluation parameters is shown in the following figure, which uses normalized values.

The radar chart in the figure below represents the comprehensive performance of the proposed system in terms of robustness, imperceptibility, security, and computational efficiency.

The proposed method has shown almost maximum performance in terms of NC robustness, which ensures the accurate recovery of watermarks. The PSNR values are maintained at higher levels, which confirm the fact that the proposed method ensures minimum distortion in the watermarked images.

The proposed method has shown robust performance in terms of noise resistance, which ensures the security of the watermarked images from signal processing attacks like Gaussian and Poisson noise. The security of the proposed method has been significantly improved by using the Hessenberg-SVD base functions.scrambling mechanism, which ensures that watermark reconstruction is heavily dependent on the secret key.

The blind extraction property provides the maximum performance, and this again points to the major advantage of the proposed method by removing the requirement for the original host image during watermark extraction. In addition to this, the computational efficiency is also high due to the use of the DCT-based embedding scheme without the inclusion of complex transformation tools such as the DWT.

The radar chart indicates that the proposed watermarking framework provides a balanced performance in all the important evaluation metrics.

VI. Conclusion and Future Enhancements

A strong and secure method of blind watermarking for confirming identity of color images has been developed in this work through the use of the Discrete Cosine Transform (DCT) and a scrambling technique based on a decomposition of the Hessenberg-Singular Value Decomposition (SVD). The proposed approach uses both frequency-domain embedding techniques and algebraically based transformation methods to achieve good security and robust performance. The watermarked signal is created by embedding watermark information into the mid-frequency DCT coefficients using a modulation technique that produces both the imperceptibility of the original image and resistance to common types of processing attacks (such as adding noise, compressing the image, or filtering it). In addition, the added level of security provided by the use of the Hessenberg-SVD based scrambling technique provides a unique way to generate content-dependent permutations, which helps prevent unauthorized extraction of the watermark. Results from the experiments show that the system provides high PSNR values (indicating little visual distortion) and NC values close to one (indicating the watermark was successfully recovered). Furthermore, the ability to extract the watermark without having access to the original images makes this system very practical for use in any real-world multimedia application. This entire framework is an efficient, secure, and scalable digital image copyright protection and authentication mechanism.

The findings of this study indicate that there are several future research avenues available based on the limitations in the existing literature on this subject. The approach suggested here is exceptionally responsive to geometric manipulation (i.e., rotating, resizing, or cropping an image), all of which inform the accuracy of watermark extraction. As a means of augmenting robustness against geometric attacks, one future

direction may involve using geometric invariants, e.g., SIFT or SURF algorithms.

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